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The asymptotic null distribution of the Box–Pierce Q-statistic for random variables with infinite variance An application to German stock returns

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Abstract

The asymptotic null distribution of the Box–Pierce Q-statistic is considered. For random variables with probability distribution in the domain of attraction of a stable law with characteristic exponent α, $1<\alpha<2$, it is shown that this does not tend to χ^2 in distribution but, after suitable normalisation, to a rather complicated limit law. The density and selected quantiles of the limit distribution are obtained with numerical methods. In a Monte Carlo experiment the small sample behaviour of the modified test is studied and an empirical application to the 30 most busy German stock returns shows that the significance of observed higher-order correlations is thereby reduced in general.

Key words: Asymptotic distribution; Autocorrelation; Domain of attraction; Portmanteau test; Stable law
JEL classification: C12

1. The problem

The Box–Pierce test (1970) has long been a favourite tool to test for zero autocorrelation in economic time series. It is based on the statistic

$$\tilde{Q}_{n,m}=n\sum_{\tau=1}^{m}\tilde{r}_\tau^2, \quad m<n, \tag{1}$$

where n is large relative to m and

$$\tilde{r}_\tau=\frac{\sum_{i=\tau+1}^{n}(e_i-\bar{e})(e_{i-\tau}-\bar{e})}{\sum_{i=1}^{n}(e_i-\bar{e})^2}, \quad \tau=1,\ldots,m$$

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is the autocorrelation of order τ of some time series $\{e_t\}$. Initially, $\{e_t\}$ were supposed to be residuals from an ARMA(p,q) process, but the test can, of course, be applied to other data as well.

Box and Pierce (1970) note that if the residuals are white noise and the model is correct, then for moderately large $m > p+q$ the statistic $\tilde{Q}_{n,m}$ in (1) is asymptotically χ^2_{m-p-q} and the model is rejected if $\tilde{Q}_{n,m} > \chi^2_{m-p-q;\,1-\theta}$, where $\chi^2_{m-p-q;\,1-\theta}$ is the $1-\theta$ quantile of a χ^2-distribution with $m-p-q$ degrees of freedom.

Ljung and Box (1978) suggest a test which is asymptotically equivalent to (1), based on

$$\tilde{Q}^*_{n,m} = n(n+2)\sum_{\tau=1}^{m} \frac{1}{n-\tau}\tilde{r}^2_\tau, \quad m < n. \tag{2}$$

They show that $\tilde{Q}^*_{n,m}$ has a higher rate of convergence to the χ^2_{m-p-q}-distribution, as confirmed by Ansley and Newbold (1979).

Now let $X_1, X_2, \ldots$ be a sequence of identically distributed random variables and consider the problem of testing the hypothesis of zero autocorrelation up to order $m, m < n$, i.e.

$$\mathrm{H}_0\colon \rho_1 = \rho_2 = \cdots = \rho_m = 0$$

against

$$\mathrm{H}_1\colon \rho_\tau \neq 0 \quad \text{for at least one } \tau,\ \tau = 1,\ldots,m, \tag{3}$$

where ρ_τ is the theoretical order τ autocorrelation coefficient. The hypothesis in (3) can be tested with the Box–Pierce or Ljung–Box statistic by using

$$r_\tau = \frac{\sum_{i=\tau+1}^{n}(X_i - \bar{X})(X_{i-\tau} - \bar{X})}{\sum_{i=1}^{n}(X_i - \bar{X})^2} \tag{4}$$

instead of $\tilde{r}_\tau$, i.e.

$$Q_{n,m} = n\sum_{\tau=1}^{m} r^2_\tau. \tag{5}$$

Since here asymptotic results are of interest, in the following only the Box–Pierce statistic is considered. Because of the asymptotic equivalence the same results hold for the Ljung–Box statistic.

If the X_i's are independently and identically distributed with $\mathrm{E}(X_i^2) < \infty$, we have

$$(\sqrt{n}\,r_1, \ldots, \sqrt{n}\,r_m) \xrightarrow{\mathrm{d}} \mathcal{N}(0_m, \mathrm{I}_m) \tag{6}$$

as $n \to \infty$ (Anderson, 1971, chap. 8), where $\xrightarrow{\mathrm{d}}$ denotes convergence in distribution, 0_m is the m-dimensional zero vector and I_m is the $(m \times m)$-identity matrix.

This implies that

$$Q_{n,m} \xrightarrow{\mathrm{d}} \chi^2_m \quad \text{as } n \to \infty. \tag{7}$$

If we allow X_i to have infinite variance, but $\mathrm{E}(X_i) < \infty \ \forall i$, then for $\tau \leqslant m$

$$\sqrt{n}\, r_\tau \xrightarrow{\mathrm{P}} 0 \quad \text{as } n \to \infty$$

(Krämer and Runde, 1994), which implies

$$Q_{n,m} \xrightarrow{\mathrm{P}} 0 \quad \text{as } n \to \infty,$$

where $\xrightarrow{\mathrm{P}}$ denotes convergence in probability. This shows that $\sqrt{n}$ is no suitable norming constant for r_τ in the infinite variance case and an asymptotically larger factor is required for $Q_{n,m}$ to have a nondegenerate limit.

2. The limiting null distribution of the Box–Pierce Q-statistic under infinite variance

In the following, we consider a sequence of identically distributed random variables $X_1, X_2, \ldots$ of the Pareto–Lévy form, i.e.

$$\lim_{x \to -\infty} \mathrm{P}(X_i < x) = c_1 \gamma^\alpha |x|^{-\alpha}[1 + \mathrm{o}(1)], \quad x < 0, \tag{8}$$

$$\lim_{x \to \infty} \mathrm{P}(X_i > x) = c_2 \gamma^\alpha x^{-\alpha}[1 + \mathrm{o}(1)], \quad x > 0, \tag{9}$$

where $1 < \alpha < 2, \gamma$ is a scale parameter and $c_1, c_2 \geqslant 0$ with $c_1 + c_2 > 0$. Note that (8) and (9) are necessary and sufficient conditions for X_i to belong to the normal domain of attraction of a stable law with characteristic exponent α (Ibragimov and Linnik, 1971, p. 76ff). Of course, stable random variables themselves follow (8) and (9) and therefore they lie in their own domain of attraction.

We denote a stable random variable with location parameter zero and scale parameter one by $S(\alpha, \beta)$. It has the characteristic function (Ibragimov and Linnik, 1971, p. 43)

$$\varphi_{S(\alpha,\beta)}(t) = \exp\{|t|^\alpha[1 - \mathrm{i}\beta\, \mathrm{sign}(t) \tan(\tfrac{\pi\alpha}{2})]\}, \tag{10}$$

where $0 < \alpha < 2, \alpha \neq 1$, is the characteristic exponent and $\beta = (c_1 - c_2)/(c_1 + c_2)$, $|\beta| \leqslant 1$, is the skewness parameter of the stable distribution. For $c_1 = c_2$, i.e. $\beta = 0$ the distribution is symmetric. It is well known that for random variables X_i of the Pareto–Lévy form $\mathrm{E}(X_i^\rho) = \infty$ for $\rho \geqslant \alpha$, i.e. $\mathrm{Var}(X_i) = \infty$ for $\alpha < 2$.

Theorem. Let $X_1, X_2, \ldots$ be a sequence of independent, identically distributed random variables which belong to the domain of attraction of a symmetric

stable distribution with characteristic exponent α, $1<\alpha<2$. *Then, as* $n \to \infty$,

$$\left(\frac{n}{\ln(n)}\right)^{2/\alpha} \sum_{\tau=1}^{m} r_\tau^2 \xrightarrow{d} Q_m(\alpha), \tag{11}$$

where

$$Q_m(\alpha) \stackrel{d}{=} \frac{1}{S^2(\frac{\alpha}{2},1)}(S_1^2(\alpha,0)+\cdots+S_m^2(\alpha,0)), \tag{12}$$

and $\stackrel{d}{=}$ *denotes equality in distribution.*

In addition, $S^2(\alpha/2,1), S_1^2(\alpha,0),\ldots,S_m^2(\alpha,0)$ *on the right-hand side of* (12) *are independent.*

Proof. Following Davis and Resnick (1986), we have for some constant $c>0$, $1<\alpha<2$ and $m \in \mathbb{N}$, $m<n$,

$$\begin{aligned}&\Bigg((c^\alpha n)^{-2/\alpha}\sum_{i=1}^{n}(X_i-\bar{X})^2,\ [c^{2\alpha}n\ln(n)]^{-1/\alpha}\sum_{i=2}^{n}(X_i-\bar{X})(X_{i-1}-\bar{X}),\ldots,\\ &\quad [c^{2\alpha}n\ln(n)]^{-1/\alpha}\sum_{i=m+1}^{n}(X_i-\bar{X})(X_{i-m}-\bar{X})\Bigg)\\ &\quad \xrightarrow{d}(S(\tfrac{\alpha}{2},1),S_1(\alpha,0),\ldots,S_m(\alpha,0)),\end{aligned} \tag{13}$$

where $S(\frac{\alpha}{2},1)$, $S_1(\alpha,0),\ldots,S_m(\alpha,0)$ in (13) are independent.

Since for random variables Y_i, Y_{ij}, $i=1,\ldots,k$, $j=1,2,\ldots$ and a continuous function $g:\mathbb{R}^k \to \mathbb{R}^l$ (see Serfling, 1980, p. 24):

$$\begin{aligned}&(Y_{1n},\ldots,Y_{kn}) \xrightarrow{d} (Y_1,\ldots,Y_k) \qquad \text{as } n\to\infty\\ &\Rightarrow g(Y_{1n},\ldots,Y_{kn}) \xrightarrow{d} g(Y_1,\ldots,Y_k) \quad \text{as } n\to\infty,\end{aligned} \tag{14}$$

with $g:\mathbb{R}^{m+1} \to \mathbb{R}_+$, $g(y_1,\ldots,y_{m+1})=(y_2^2+\cdots+y_{m+1}^2)/y_1^2$ the theorem is proved. □

A closer inspection of the ratio of the new norming constant in (11) over that in the standard case reveals that for any $1<\alpha<2$

$$\lim_{n\to\infty}\frac{[n/\ln(n)]^{2/\alpha}}{n}=\infty, \tag{15}$$

which seems to indicate that the significance of higher-order correlations with given values r_τ is now increased. But since the convergence to infinity in (15) is neither monotone in n nor uniform in α the new norming constant is smaller

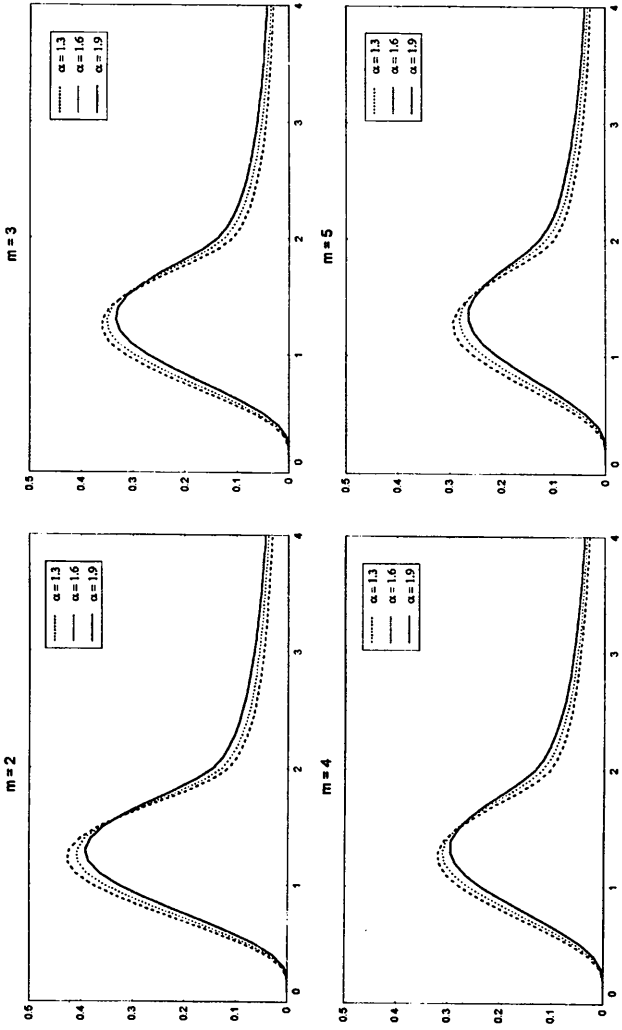

Fig. 1. Densities of $Q_m(x)$.

than the old for the most relevant cases of empirical research, say $n \leqslant 10\,000$ and $\alpha > 1.5$, which implies that the significance of the Box–Pierce test statistic is reduced. In addition, although $Q_m(\alpha)$ is continuous with respect to α at the endpoints $\alpha = 1$ and 2, it should be pointed out that the asymptotic result in (11) does not hold for $\alpha = 1$ and 2, respectively, since a stable variate with $\alpha = 1$ follows a Cauchy law and a stable variate with $\alpha = 2$ is normal.

The results in (11) and (12) are impractical as they stand, since for empirical work one needs at least critical values of the limiting distribution. This has to be done with computer, since the distribution function of $Q_m(\alpha)$ – similar to that of a stable variate itself – is not available in closed form for $1 < \alpha < 2$. Therefore, Bergström's (1952) well-known series expansions and their approximations for the distribution functions of stable variates are used to compute the values for the densities and some critical values of the limiting distribution $Q_m(\alpha)$ numerically (for details see Runde, 1993). The results are given in Fig. 1 and Table 1 for selected values of m and α.

Table 1
Critical values of the limiting distribution $Q_m(\alpha)$

	$m = 2$				$m = 3$		
	$1 - \theta$				$1 - \theta$		
α	**0.90**	**0.95**	**0.99**	α	**0.90**	**0.95**	**0.99**
1.90	15.60	17.94	22.07	**1.90**	17.95	20.43	23.17
1.80	14.29	16.74	20.82	**1.80**	16.79	19.46	21.92
1.70	13.41	15.83	19.68	**1.70**	15.74	18.21	20.84
1.60	12.09	14.47	18.54	**1.60**	14.63	17.14	19.63
1.50	11.12	13.53	17.38	**1.50**	13.58	16.32	18.49
1.40	10.27	12.37	16.20	**1.40**	12.50	15.10	17.41
1.30	9.31	11.52	15.03	**1.30**	11.77	13.97	16.33
1.20	8.87	10.87	13.92	**1.20**	10.94	13.00	15.59
1.10	8.11	10.05	13.04	**1.10**	10.02	11.88	14.70
	$m = 4$				$m = 5$		
	$1 - \theta$				$1 - \theta$		
α	**0.90**	**0.95**	**0.99**	α	**0.90**	**0.95**	**0.99**
1.90	20.25	23.45	25.56	**1.90**	21.75	24.87	26.31
1.80	19.13	22.01	24.13	**1.80**	20.57	23.16	25.14
1.70	17.95	20.75	22.99	**1.70**	19.42	21.73	24.08
1.60	17.00	19.43	21.90	**1.60**	18.23	20.43	22.95
1.50	15.98	18.28	20.79	**1.50**	17.20	19.17	21.89
1.40	14.78	17.30	19.75	**1.40**	16.11	18.24	21.06
1.30	13.95	16.25	18.64	**1.30**	15.09	17.14	19.97
1.20	13.11	15.14	17.77	**1.20**	14.28	15.92	19.04
1.10	12.44	14.31	16.83	**1.10**	13.70	15.08	18.18

Table 2
Empirical significance levels (in percentage)

		$n = 10$			$n = 20$				$n = 50$				
		m			m				m				
% level	α	2	3	5	2	3	5	10	2	3	5	10	20
	1.3	4.8	3.9	3.5	3.5	4.2	3.9	3.8	4.1	4.2	3.7	4.1	5.8
5	1.6	4.2	2.9	3.1	2.8	3.3	2.9	3.1	5.2	4.3	4.4	3.7	3.9
	1.9	3.3	3.7	3.0	3.1	2.2	2.8	2.5	3.5	3.8	2.5	2.0	3.2
	1.3	9.1	10.9	8.3	9.8	11.1	8.2	8.9	8.6	8.2	9.0	10.4	11.3
10	1.6	8.6	7.8	8.0	11.4	10.9	9.1	7.6	8.0	7.1	6.9	12.6	8.2
	1.9	8.1	7.6	5.9	8.2	8.0	13.3	7.5	12.3	11.6	8.2	8.4	6.5
	1.3	18.2	18.4	17.5	19.2	18.6	17.8	18.1	17.4	19.0	18.2	22.5	17.9
20	1.6	23.6	20.6	24.2	20.8	21.5	17.3	16.9	18.7	16.8	17.7	16.5	16.2
	1.9	17.4	17.2	13.2	16.3	14.1	15.7	10.3	17.3	15.6	16.4	14.8	11.6

3. Monte Carlo results

The Monte Carlo experiment in this Section focuses on the small sample performance of the modified Box–Pierce test. To investigate this issue sample sizes of $n = 10, 20$ and 50 were considered and for each series length 10 000 independent realisations of stable variates with characteristic exponent $\alpha = 1.3, 1.6$ and 1.9, respectively, were generated using the algorithm proposed by Chambers et al. (1976). For each realisation the modified statistic (11) was computed for several values of m. To appreciate the accuracy of the $Q_m(\alpha)$ approximation, the empirical frequencies of rejection of the null hypothesis of serial independence was examined by tests with three different nominal levels of $\theta = 0.05, 0.1$ and 0.2.

The results in Table 2 show that throughout the whole experiment the $Q_m(\alpha)$ distribution provides a relatively poor approximation for small sample sizes. In most cases, the empirical levels are lower than the theoretical ones and the worst results are obtained for α close to two and m large relative to n.

4. Empirical application

Since Mandelbrot (1963) and Fama (1965) stock returns $S_t := \ln(K_t/K_{t-1})$, where K_t is the price of some common stock at the end of period t, are often assumed to follow stable laws. This section reports an empirical application of the above results for the returns of German stocks. We include all stocks that make up the DAX, except Lufthansa, Henkel, Veba and Viag, for which no uninterrupted series of returns could be obtained. Time ranges from January 4th, 1960 until January 31st, 1992, comprising $n = 8026$ trading days on the Frankfurt stock exchange. The data was provided by the Deutsche Finanzdatenbank (DFDB) in Karlsruhe.

First, we test whether the assumptions of the theorem hold, i.e. whether the stock returns belong to the domain of attraction of a symmetric stable distribution with characteristic exponent α, $1<\alpha<2$. This is given if

(a) the distribution of the returns itself is symmetric and

(b) the tail probability of the distribution $1-F(x)$ is proportional to $1/x^{\alpha}$ for $x \to \infty$ with $1<\alpha<2$.

The symmetry is tested with the Wilcoxon signed rank test (WT), which is asymptotically normal. The realisations of the Wilcoxon statistic are listed in Table 3. It shows that for none of the considered stocks the hypothesis of symmetry can be rejected at a significance of 5%.

With a consistent estimation of α we can investegate whether (b) is true or not. The procedure that leads to the estimators given in Table 3 is another nonparametric method based on Hill (1975). The asterisks in Table 3 identify the stocks with $1<\hat{\alpha}<2$.

The empirical results of the Box–Pierce test for the stocks with an estimated α less than 2 are summerized in Table 4 for $m=2,\ldots,5$. The first column gives the standard Box–Pierce statistic. It rejects the null hypothesis of serial independence for all stocks at a significance of less than 1%, indicated by an asterisk.

The second column reports the realisations of a nonparametric portmanteau rank test

$$Q^*_{n,m} = \sum_{\tau=1}^{m} \frac{(r^*_\tau - \mu_\tau)^2}{\sigma^2_\tau}$$

Table 3
Wilcoxon signed rank test statistic and estimates of α

	WT	$\hat{\alpha}$		WT	$\hat{\alpha}$
BASF*	1.342	1.84	RWE	1.105	2.82
BMW*	0.980	1.74	Schering*	0.972	1.92
Continental*	1.131	1.81	Siemens*	0.933	1.83
Daimler*	0.624	1.74	Thyssen*	0.892	1.91
Deutsche Babcock	0.874	2.57	VW*	1.243	1.80
Degussa*	1.116	1.88	Kaufhof*	1.193	1.85
Bayer*	1.028	1.82	Bayr. Hypo.	0.931	2.79
Hoechst*	1.028	1.85	Bayr. Vereinsbank	0.645	2.32
MAN*	0.979	1.84	Commerzbank*	0.773	1.89
Karstadt*	0.623	1.91	Deutsche Bank*	1.088	1.80
Linde	1.225	2.65	Dresdner Bank*	0.935	1.81
Mannesmann	1.389	2.83	Allianz	1.217	2.61
Metallgesellschaft	0.762	2.74	DAX	0.599	2.71
Preussag*	0.894	1.87			

* Stocks with $1<\hat{\alpha}<2$.

Table 4
Empirical results of the Box–Pierce test for German stock returns

Company	$n\sum r_\tau^2$	$\sum \frac{(r_\tau^* - \mu_\tau)^2}{\sigma_\tau^2}$ $m = 2$	$(\frac{n}{\ln(n)})^2 \doteq \sum r_\tau^2$	$n\sum r_\tau^2$	$\sum \frac{(r_\tau^* - \mu_\tau)^2}{\sigma_\tau^2}$ $m = 3$	$(\frac{n}{\ln(n)})^2 \doteq \sum r_\tau^2$
BASF	61.03*	54.02*	12.26	62.59*	55.37*	12.57
BMW	148.12*	103.90*	45.47*	177.13*	105.76*	54.37*
Continental	32.49*	23.04*	7.38	39.93*	23.27*	9.06
Daimler	138.72*	90.02*	42.58*	145.49*	90.11*	44.66*
Degussa	25.87*	24.28*	4.44	26.95*	24.32*	4.62
Bayer	34.75*	20.28*	7.57	34.76*	20.29*	7.57
Hoechst	63.87*	44.40*	12.32	64.11*	45.23*	12.37
MAN	44.28*	28.98*	8.89	44.39*	30.43*	8.92
Karstadt	16.22*	14.43*	2.49	16.69*	14.48*	2.59
Preussag	34.57*	25.25*	6.17	36.02*	25.51*	6.43
Schering	49.24*	44.06*	7.27	49.28*	44.13*	7.28
Siemens	49.48*	31.79*	10.35	57.61*	36.15*	12.05
Thyssen	61.84*	50.20*	9.48	63.37*	52.68*	9.71
VW	28.39*	20.84*	6.76	28.61*	21.30*	6.82
Kaufhof	51.85*	43.35*	10.01	52.32*	43.47*	10.09
Commerzbank	75.05*	54.58*	12.40	78.22*	59.29*	12.92
Deutsche Bank	110.57*	99.12*	26.16*	117.95*	106.71*	27.91*
Dresdner Bank	128.27*	106.94*	29.11*	128.29*	106.99*	29.12*

Table 4. Continued.

Company	$n\sum r_\tau^2$	$\sum \frac{(r_\tau^* - \mu_\tau)^2}{\sigma_\tau^2}$ $m = 4$	$(\frac{n}{\ln(n)})^2 \hat{\alpha} \sum r_\tau^2$	$n\sum r_\tau^2$	$\sum \frac{(r_\tau^* - \mu_\tau)^2}{\sigma_\tau^2}$ $m = 5$	$(\frac{n}{\ln(n)})^2 \hat{\alpha} \sum r_\tau^2$
BASF	64.17*	60.94*	12.89	65.62*	62.08*	13.18
BMW	177.36*	107.31*	54.44*	182.09*	107.46*	55.89*
Continental	41.34*	24.29*	9.38	41.44*	26.16*	9.40
Daimler	152.54*	96.29*	46.82*	152.69*	96.31*	46.87*
Degussa	27.53*	25.74*	4.73	27.55*	25.80*	4.73
Bayer	34.76*	21.36*	7.57	35.17*	22.64*	7.66
Hoechst	64.69*	45.90*	12.48	66.23*	45.96*	12.78
MAN	44.78*	33.28*	8.99	44.83*	36.19*	9.01
Karstadt	19.10*	14.59*	2.93	19.62*	15.31*	3.01
Preussag	37.21*	28.63*	6.64	37.49*	28.96*	6.69
Schering	56.88*	47.95*	8.39	57.43*	48.05*	8.48
Siemens	64.38*	43.99*	13.46	64.48*	44.29*	13.48
Thyssen	65.44*	53.37*	10.03	65.67*	53.89*	10.06
VW	29.90*	23.76*	7.12	30.10*	23.86*	7.17
Kaufhof	52.84*	44.17*	10.15	55.53*	44.25*	10.72
Commerzbank	82.28*	74.11*	13.59	82.33*	77.27*	13.60
Deutsche Bank	129.04*	118.77*	30.54*	129.26*	118.79*	30.59*
Dresdner Bank	139.49*	115.60*	31.66*	140.26*	115.63*	31.84*

* Significance level less than 1%.

proposed by Dufour and Roy (1986). Here r_τ^* is the rank autocorrelation at lag τ where each return S_i is replaced by its rank R_i

$$r_\tau^* = \frac{\sum_{i=\tau+1}^{n}(R_i - \bar{R})(R_{i-\tau} - \bar{R})}{\sum_{i=1}^{n}(R_i - \bar{R})^2}, \quad 1 \leqslant \tau \leqslant n-1,$$

$$\mu_\tau = \frac{-(n-\tau)}{n(n-1)}, \quad 1 \leqslant \tau \leqslant n-1,$$

and

$$\sigma_\tau^2 = \frac{5n^4 - (5\tau+9)n^3 + 9(\tau-2)n^2 + 2\tau(5\tau+8)n + 16\tau^2}{5(n-1)^2 n^2 (n+1)}, \quad 1 \leqslant \tau \leqslant \frac{n}{2}.$$

Independent of the shape of the underlying data distribution, $Q_{n,m}^*$ is asymptotically χ_m^2 (Dufour and Roy, 1986). The realisations of the rank portmanteau statistic provide less significant results than the standard Box–Pierce statistic, but this nonparametric procedure still rejects the null hypothesis for all stocks at a significance of less than 1%.

Table 4 also presents the results of the Box–Pierce test if we allow the returns to belong to the domain of attraction of stable distributions. Since for a continuous function g and a consistent estimator $\hat{\alpha}$ of the unknown parameter $g(\hat{\alpha}) - g(\alpha) \xrightarrow{P} 0$, the test statistic in (11) has the same limit distribution when α is replaced by a consistent estimate. Hence, the asymptotic relation in (11) is still valid, if we replace α by the Hill estimator, which is consistent (Mason, 1982). However, we see that for almost all stocks the significance, if judged by the new asymptotic theory, is reduced. Except BMW, Daimler, Deutsche Bank and Dresdner Bank, which are still significant at 1% for all other stocks the null hypothesis of serial independence cannot be rejected even with a significance level of 10%.

References

Anderson, T.W., 1971, The statistical analysis of time series (Wiley, New York).

Ansley, C.F. and P. Newbold, 1979, On the finite sample distribution of residual autocorrelations in autoregressive-moving average models, Biometrika 66, 547–553.

Bergström, H., 1952, On some expansions of stable distribution functions, Arkiv för Matematik, 2, 375–378.

Box, G.E.P. and D.A. Pierce, 1970, Distribution of residual autocorrelations in autoregressive-integrated moving average time series models, Journal of the American Statistical Association 65, 1509–1526.

Chambers, J.M., C.L. Mallows and B.W. Stuck, 1976, A method for simulating stable random variables, Journal of the American Statistical Association 71, 340–344.

Davis, R. and Resnick, S., 1986, Limit theory for the sample covariance and correlation functions of moving averages, The Annals of Statistics 14, 533–558.

Dufour, J.-M. and R. Roy, 1986, Generalized portmanteau statistics and tests of randomness, Communications in Statistics, Theory and Methods 15, 584–595.

Fama, E., 1965, The behaviour of stock market prices, Journal of Business 38, 34–105.
Hill, B.M., 1975, A simple general approach to inference about the tail of a distribution, The Annals of Statistics 3, 1163–1174.
Ibragimov, I.A. and Y.V. Linnik, 1971, Independent and stationary sequences of random variables (Wolters-Noordhoff, Groningen).
Krämer, W. and R. Runde, 1994, Some pitfalls in using empirical autocorrelations to test for zero correlation among common stock returns, in: Kaehler, J. and Kugler, P., eds., Econometric analysis of financial markets (Physica-Verlag, Heidelberg) 1–10.
Ljung, G.M. and G.E.P. Box, 1978, On a measure of lack of fit in time series models, Biometrika 65, 297–303.
Mandelbrot, B., 1963, The variation of certain speculative prices, Journal of Business 36, 394–419.
Mason, D.M., 1982, Laws of large numbers for sums of extreme values, The Annals of Probability 10, 754–764.
Runde, R., 1993, Consequences of an infinite population variance for the asymptotic null distribution of selected tests on dispersion, location and autocorrelation (in German), Ph.D. Dissertation (Department of Statistics, University of Dortmund).
Serfling, R.J., 1980, Approximation theorems of mathematical statistics (Wiley, New York).